BT503
paper
Bt503 Mcqs
There are different causes of the industrial pollution in the Karachi which are given below:
There are no proper system to discard the waste of the industries.
Their is no system for cleaning up of the smoke.
There is no proper check and balance for the industries.
There is no system for the cleaning up of the water which is used in the any industry.
The pollution which is cause due to the industries is called the industrial pollution.
Worldwide there were more than 249 herbicide resistant weedy biotypes in 47 countries
Plant transformation is a scientific approach whereby DNA from any organism is inserted into the genome of a species of interest. The inserted DNA is called a “transgene”,
Good food
Less time
Infection tolerance
Insect tolerance
transformation is the geneticalteration of a cell resulting from the direct uptake and incorporation of exogenous genetic material from its surroundings through the cell membrane.
Agrobacterium
Viral transformation
Gene gun
Electroporation
The negative impact of the non-living factors on living organisms in a specific environment is called the abiotic stress.
By pre-formed structures and the chemicals
 By the infection-induced responses for the immune system.
Use of such type of material which cause less pollution
Use the air filters in the factories
Use less toxic fuels for vehicles
Zooplankton are heterotrophic plankton.

Paper:
What is recombinant?

Recombinant is the anything which have the mixture of two hereditary materials one is of the organism and the second is of the another which

hereditary material is introduced into the body of first.

What is abiotic stress?
The negative impact of the non-living factors on living organisms in a specific environment is called the abiotic stress.

What is bioaccumulation?
The collection of substances e.g. toxins and chemicals, in an organism's body is called bioaccumulation".
Sources: substances from any source e.g. air, water, soil, could be stocked through bioaccumulation.
Substances to be gathered through bioaccumulation have following properties;

- They are preferably relentless, lipophilic and stable
- Unable to translocate throughout the body
- Have low catabolic rate
- High affinity for proteins

What is toxicity?
Toxicity is the degree to which a chemical substance or a particular mixture of substances can damage an organism.
Toxicity can refer to the effect on a whole organism, such as an animal, bacterium, or plant, as well as the effect on a substructure of the organism, such as
a cell (cytotoxicity) or an organ such as the liver (hepatotoxicity).

What is the principle of transformation of plants?

- There are number of principles which are used for the transformation of the plants which are given below:

1. Vector method:
   There are two vector methods which are given below:
   1. Agrobacterium
   2. Viral transformation
2. Vector Free method:
   There are different vector free methods which are given below:
   1. Gene gun 2. Electroporation

What is pLac z expression vector?

pLac-Z expression vectors

- Lac –Z promoter operator is in frame with lac-Z alpha fragment (the NH3 terminal part of Galactosidase gene.
- Multiple cloning sites are found in the border of NH3 end including ATG sequence.

- The presence of such restriction site sequences should not disturb the functional activity of the protein, which complements with the omega fragment of the Lac-Z produced by the bacterial cell as the complement.
- If any gene is placed in proper frame in the MCS the protein expressed will be is fused form.
- The expression of the gene can be regulated by IPTG (Isopropyl thio b-Galactoside).

Types of pollution?
There are different types of the pollution which are given below:
?Air pollution
?Water pollution
?Land pollution
?Environmental pollution
?Light pollution
?Noise pollution

Write a note on plant transformation ?
Plant transformation is a scientific approach whereby DNA from any organism is inserted into the genome of a species of interest. The inserted DNA is called a “transgene”, and the
resulting plant is said to be “transgenic”. Transgenic plants are vital for both research and agriculture.
Introduction:
In plant transformation, insert the gene of our interest. In this way we can get induce the those type of characteristics which we want in the crop.
By using the process of transformation we can get the crop of our interest.
Advantages:
There are different advantages of the plant transformation which are given below:
By using the plant transformation we can protect our crop from the pesticides.
By using the techniques of the plant transformation we can doing the changes in the fruit.
Good food
?Less time ?Infection tolerance ?Insect tolerance
Examples:
There are different examples of the plant transformation which are given below|:
?Long grain rice
?Avocado without seeds
?Disease resistance fruits and plants.

How to generate recombinant virus vaccine?

Generate a Recombinant Virus Vaccine

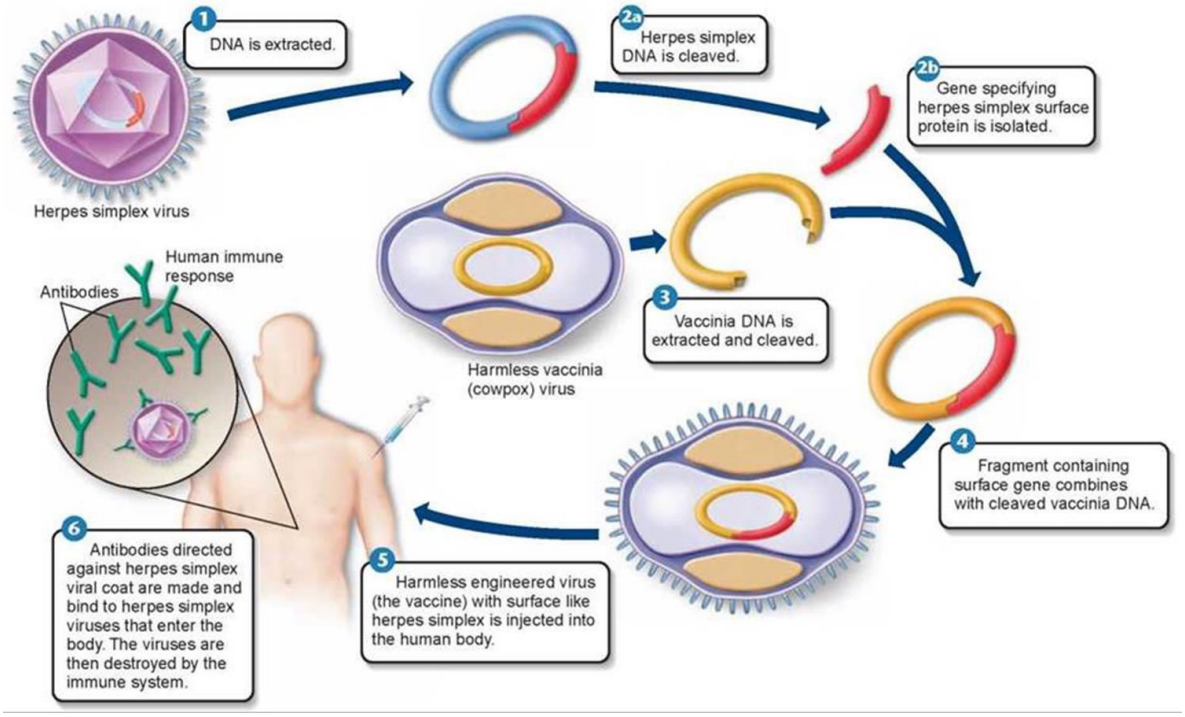
- A number of genetically stable, safe live viral vaccines have been produced, e.g. measles and yellow fever.

- Live vaccines have a number of advantages

• but when attempting to expand the range of vaccines to include new targets it is often not possible to achieve a suitable live vaccine derivative of the pathogenic virus.
• Recombinant viral vaccines offer a path to achieving the benefits of live vaccines for such difficult cases.

- The principle is to embed one or more genes from the target pathogen into the genome of an established attenuated viral vaccine strain or other nonpathogenic viral genome

- when the recombinant vaccine is administered, its genes encode proteins from the target pathogen and hence elicit protective immune responses against it.



Write a detailed note on principle of transformation of plants?Transformation principle for plants

Transformation

• transformation is the geneticalteration of a cell resulting from the direct uptake and incorporation of exogenous genetic material from its surroundings through the cell membrane.

- There are number of principles which are used for the transformation of the plants which are given below:

1. Vector method:

There are two vector methods which are given below:
1. Agrobacterium
2. Viral transformation

2. Vector Free method:
There are different vector free methods which are given below:
1. Gene gun 2. Electroporation

Vector method

• There are two vector methods which are given below:
• Agrobacterium
• Viral transformation

Agrobacterium:
Agrobacterium mediated transformation is the easiest and most simple plant transformation. Plant tissue (often leaves) are cut into small pieces, e.g. 10x10mm, and soaked for 10 minutes in a fluid containing suspended Agrobacterium. The bacteria will attach to many of the plant cells exposed by the cut. The plant cells secrete wound-related phenolic compounds which in turn act to up regulate the virulence operon of the Agrobacterium.
Viral Transformation:
Package the desired genetic material into a suitable plant virus and allow this modified virus to infect the plant. If the genetic material is DNA, it can recombine with the chromosomes to produce transforman cells. However, genomes of most plant viruses consist of single
stranded RNA which replicates in the cytoplasm of infected cell. For such genomes this method is a form
of transfection and not a real transformation, since the inserted genes never reach the nucleus of the cell and do not integrate into the host genome. The progeny of the infected plants is virus-free and also free of the inserted gene.

Vector less method

- There are two vector less method which are given below:
- Gene gun
- Electroporation
- Gene Gun:
  It is referred as particle bombardment or biolistics. Partilcles of gold or tungsten are coated with the DNA and then shot into young plant cells.
- Electroporation:
  Formation of transient holes in cell membranes using electric pulses of high field strength this allows DNA to enter as described above for bacteria.